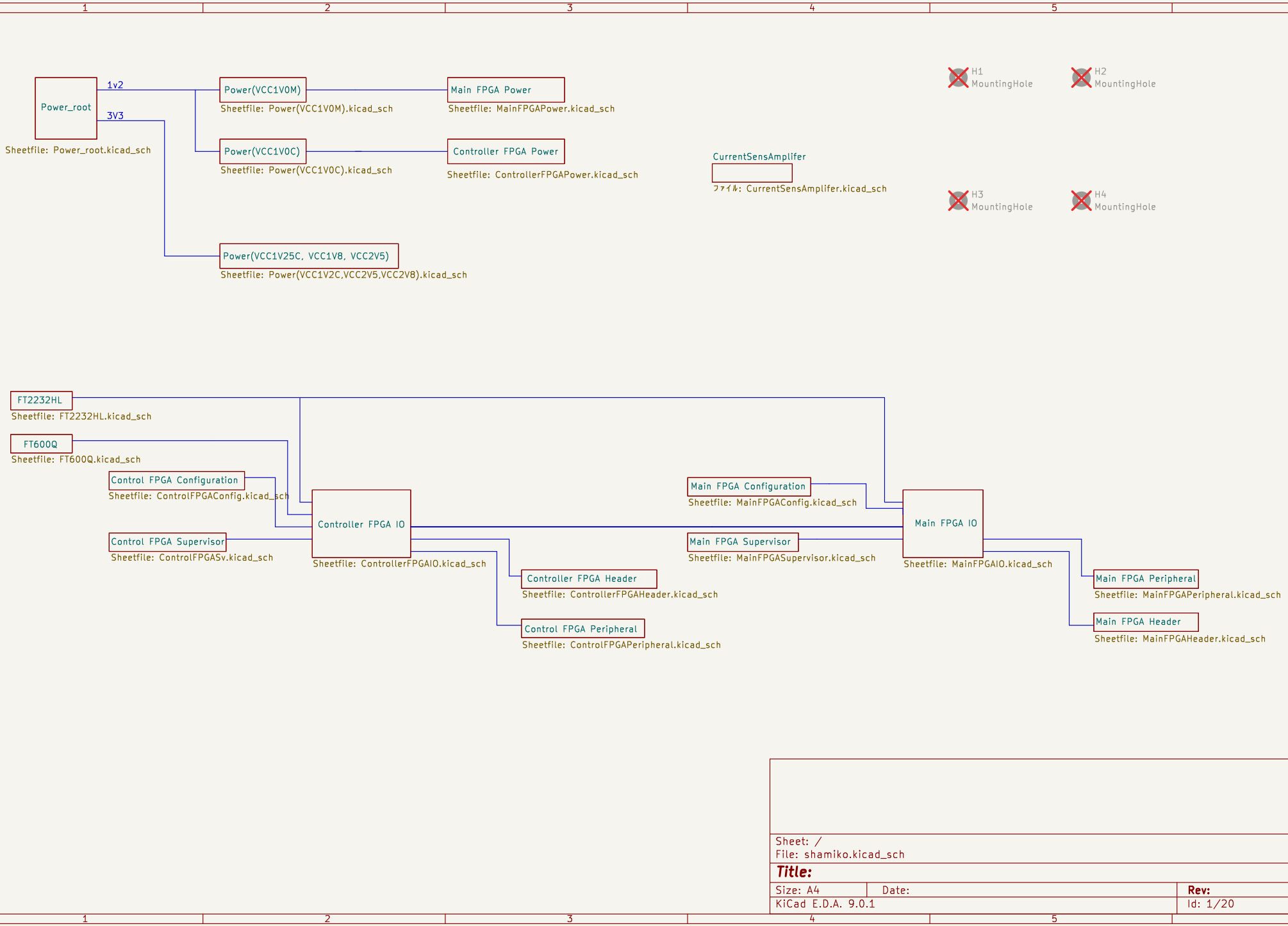
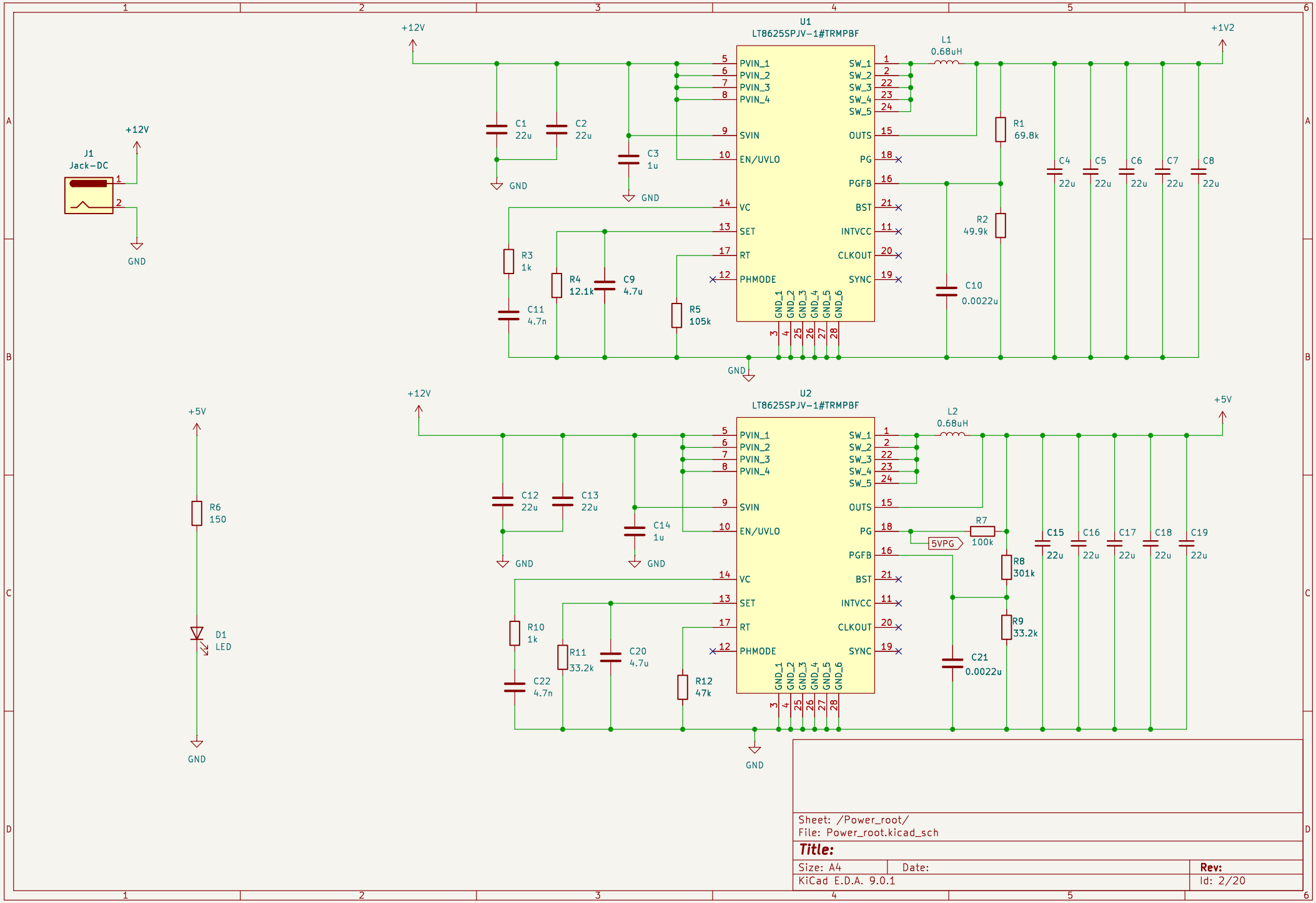
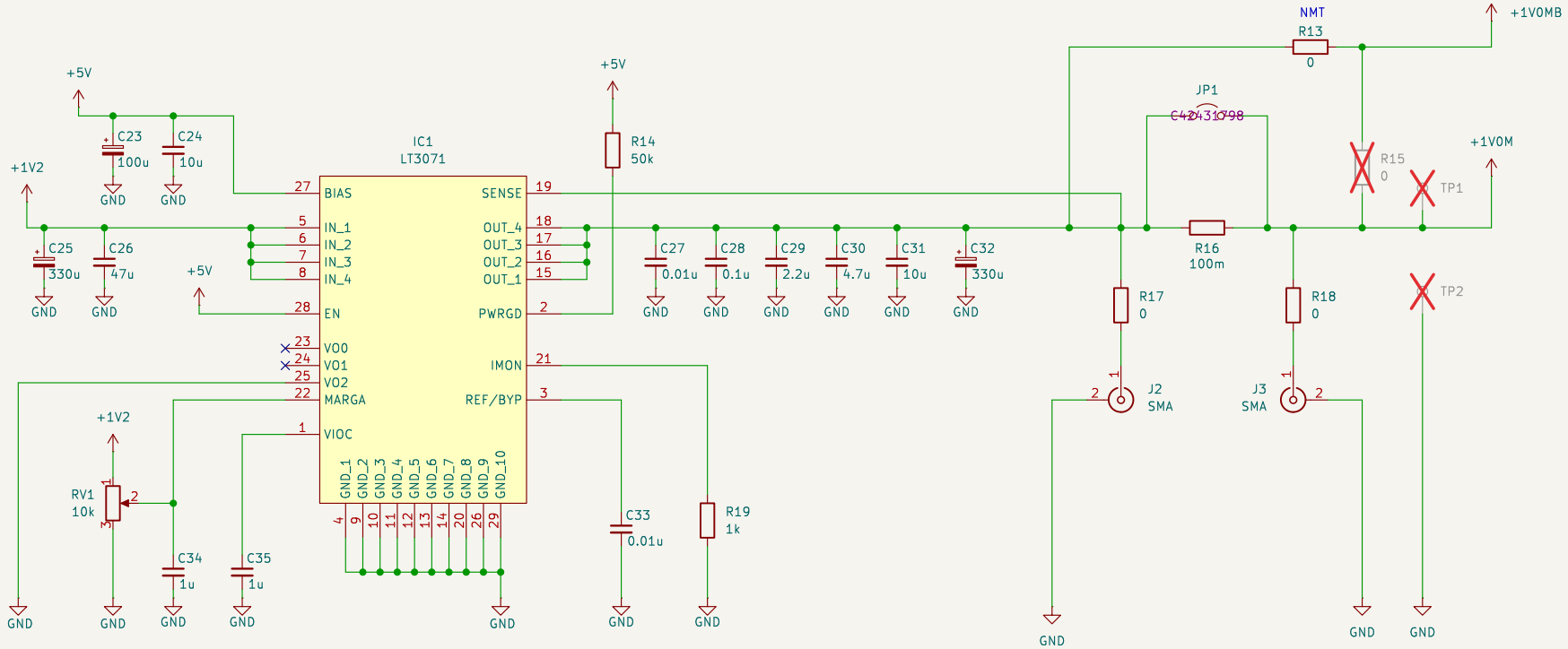
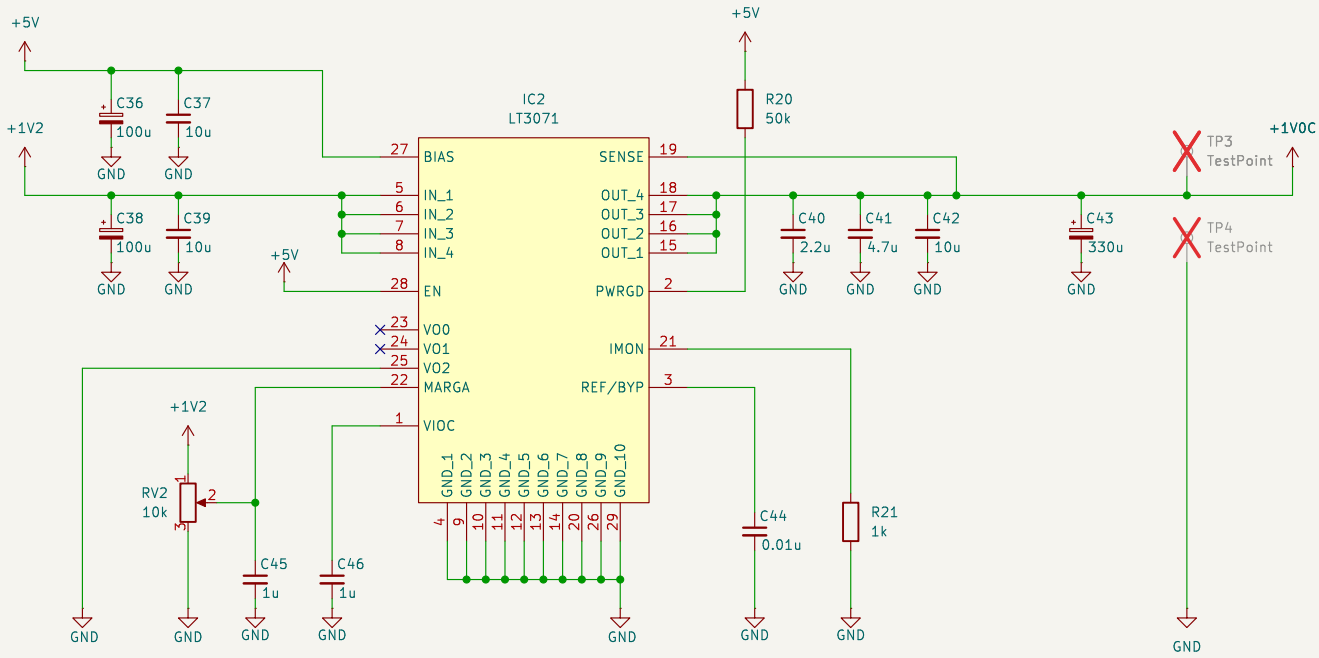

Power_root
1v2
3V3
Sheetfile: Power_root.kicad_sch
Power(VCC1V0M)
Sheetfile: Power(VCC1V0M).kicad_sch
Main FPGA Power
Sheetfile: MainFPGAPower.kicad_sch
Power(VCC1V0C)
Sheetfile: Power(VCC1V0C).kicad_sch
Controller FPGA Power
Sheetfile: ControllerFPGAPower.kicad_sch
Power(VCC1V25C, VCC1V8, VCC2V5)
Sheetfile: Power(VCC1V2C,VCC2V5,VCC2V8).kicad_sch
CurrentSensAmplifer
ファイル: CurrentSensAmplifer.kicad_sch
H1
MountingHole
H2
MountingHole
H3
MountingHole
H4
MountingHole
FT2232HL
Sheetfile: FT2232HL.kicad_sch
FT600Q
Sheetfile: FT600Q.kicad_sch
Control FPGA Configuration
Sheetfile: ControlFPGAConfig.kicad_sch
Control FPGA Supervisor
Sheetfile: ControlFPGASv.kicad_sch
Controller FPGA IO
Sheetfile: ControllerFPGAIO.kicad_sch
Controller FPGA Header
Sheetfile: ControllerFPGAHeader.kicad_sch
Control FPGA Peripheral
Sheetfile: ControlFPGAPeripheral.kicad_sch
Main FPGA Configuration
Sheetfile: MainFPGAConfig.kicad_sch
Main FPGA Supervisor
Sheetfile: MainFPGASupervisor.kicad_sch
Main FPGA IO
Sheetfile: MainFPGAIO.kicad_sch
Main FPGA Peripheral
Sheetfile: MainFPGAPeripheral.kicad_sch
Main FPGA Header
Sheetfile: MainFPGAHeader.kicad_sch
Sheet: /
File: shamiko.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.1
Id: 1/20


U1
LT8625SPJV-1#TRMPBF
U2
LT8625SPJV-1#TRMPBF
J1
Jack-DC
+12V
GND
+1V2
+5V
L1
0.68uH
L2
0.68uH
C1
22u
C2
22u
C3
1u
C4
22u
C5
22u
C6
22u
C7
22u
C8
22u
C9
4.7u
C10
0.0022u
C11
4.7n
C12
22u
C13
22u
C14
1u
C15
22u
C16
22u
C17
22u
C18
22u
C19
22u
C20
4.7u
C21
0.0022u
C22
4.7n
R1
69.8k
R2
49.9k
R3
1k
R4
12.1k
R5
105k
R6
150
R7
100k
R8
301k
R9
33.2k
R10
1k
R11
33.2k
R12
47k
5VPG
D1
LED
PVIN_1
PVIN_2
PVIN_3
PVIN_4
SVIN
EN/UVLO
VC
SET
RT
PHMODE
SW_1
SW_2
SW_3
SW_4
SW_5
OUTS
PG
PGFB
BST
INTVCC
CLKOUT
SYNC
GND_1
GND_2
GND_3
GND_4
GND_5
GND_6
Sheet: /Power_root/
File: Power_root.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.1
Id: 2/20

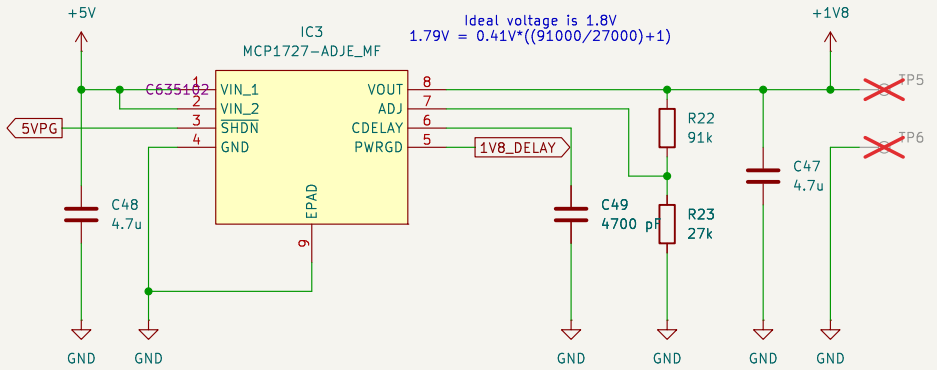


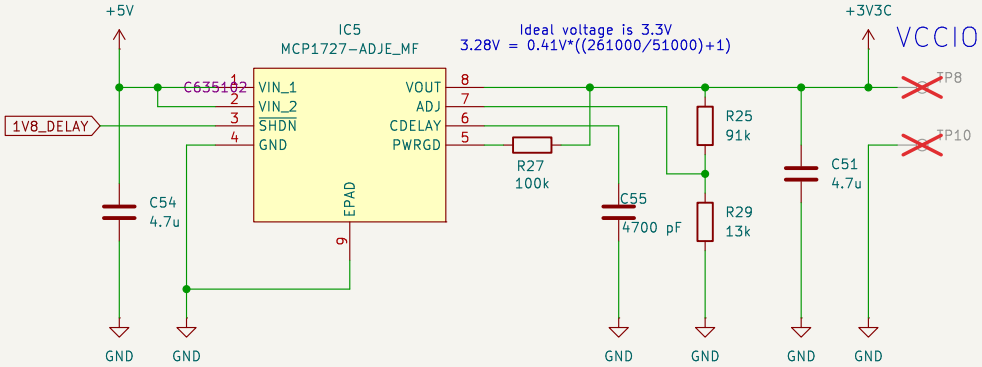
Sheet: /Power(VCC1V0M)/		
File: Power(VCC1V0M).kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.1		Id: 3/20

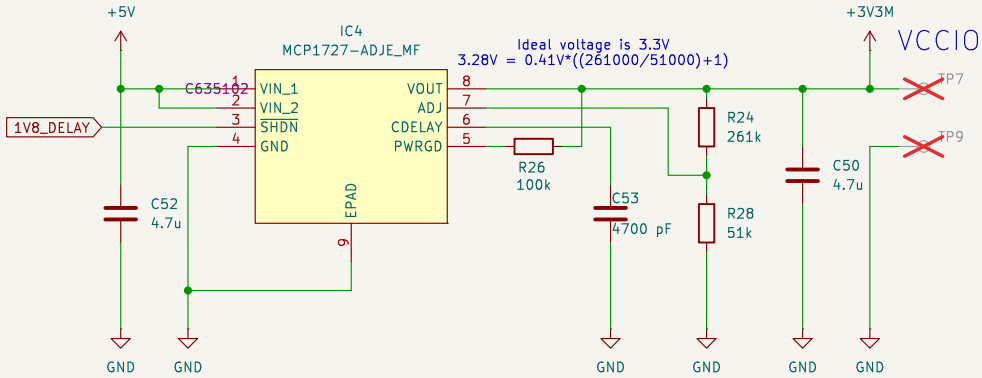


Sheet: /Power(VCC1V0C)/		
File: Power(VCC1V0C).kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.1		Id: 4/20

VCCAUX



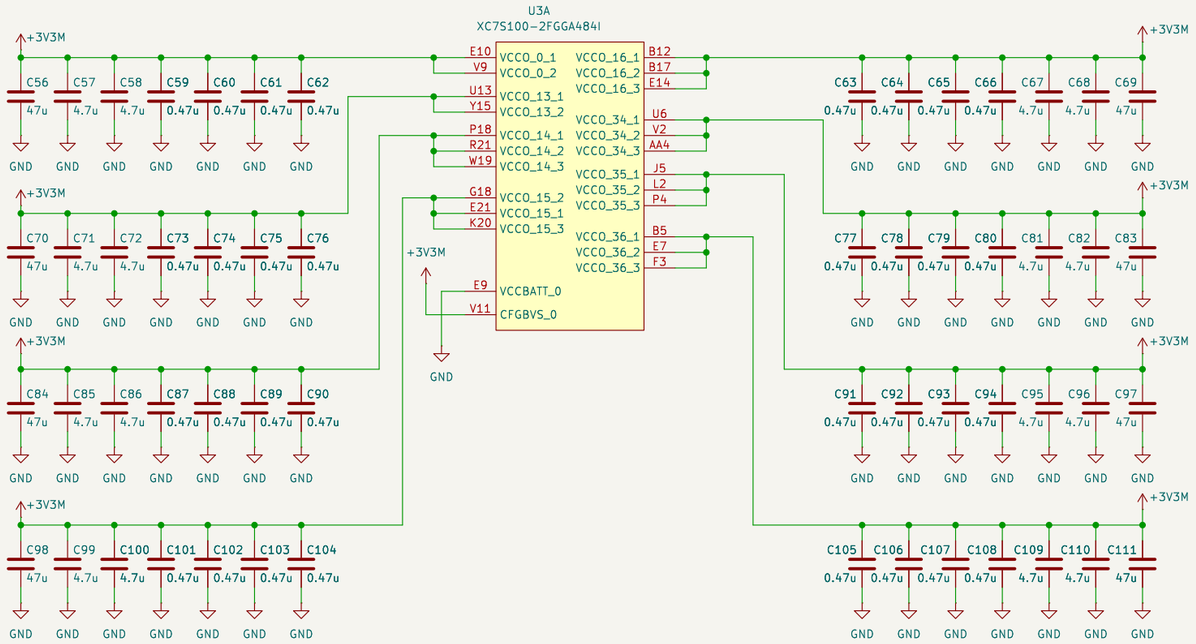


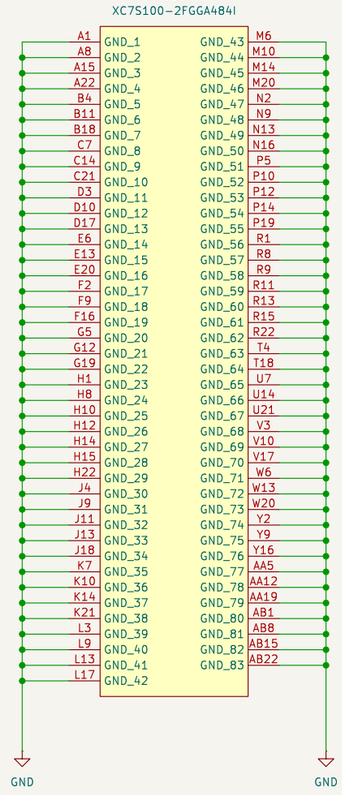


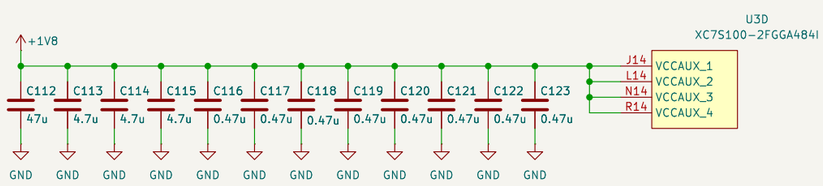
Sheet: /Power(VCC1V25C, VCC1V8, VCC2V5)/
File: Power(VCC1V2C,VCC2V5,VCC2V8).kicad_sch

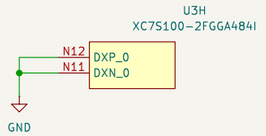
Title:

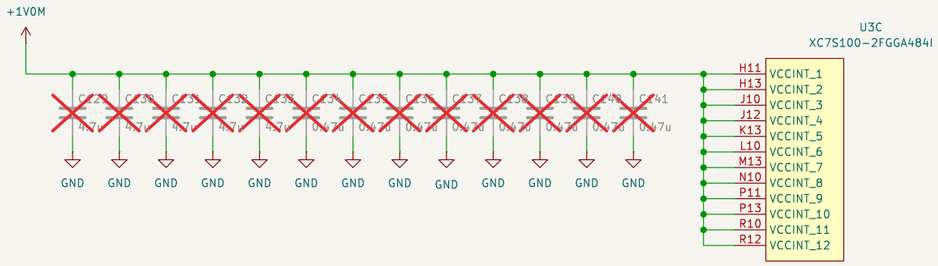
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.1		Id: 5/20

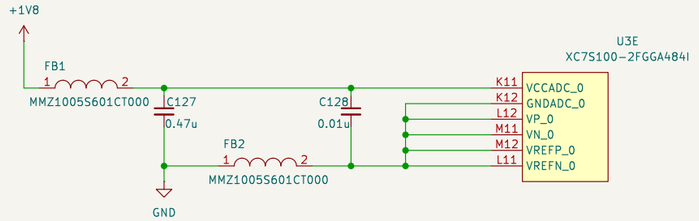




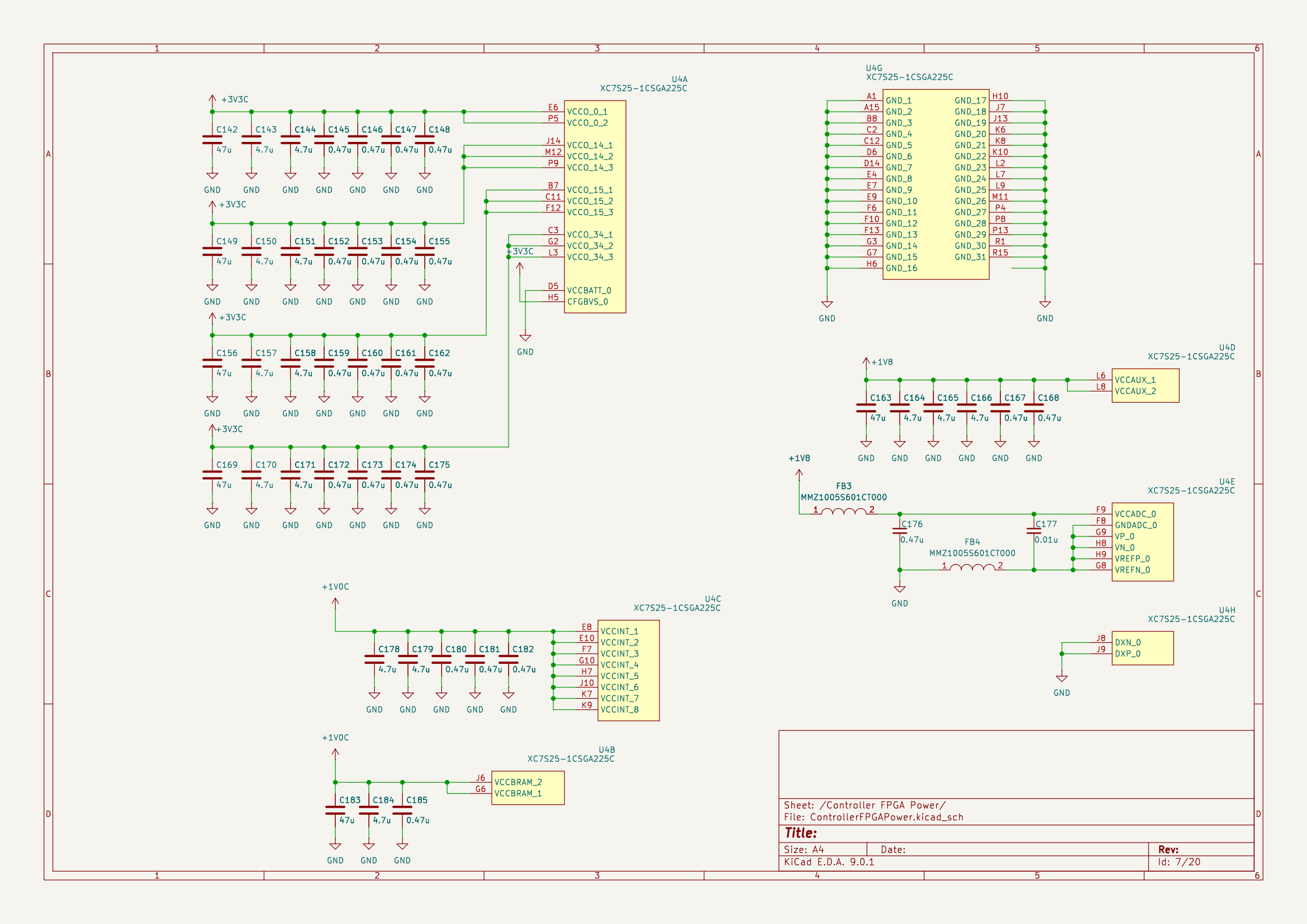


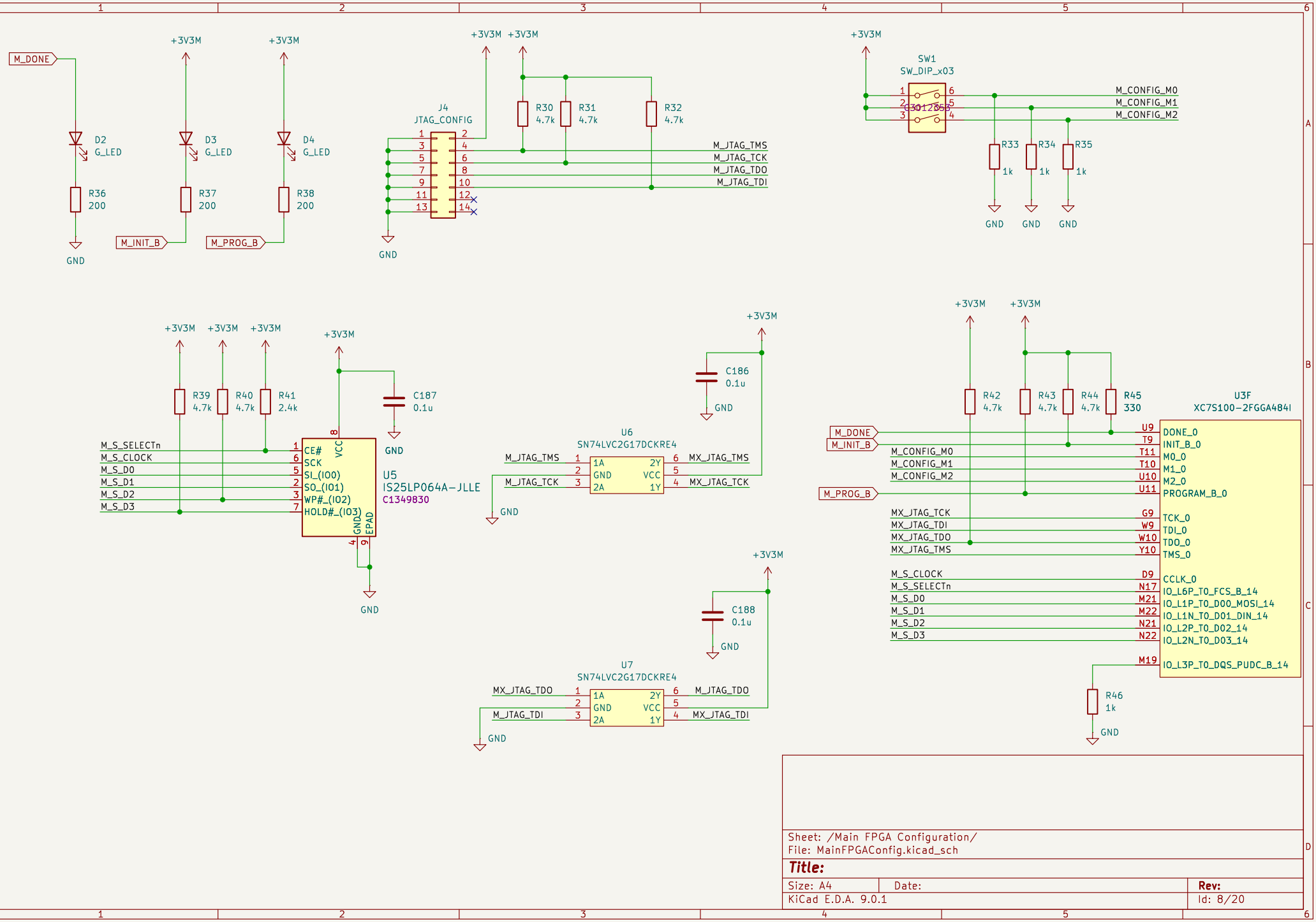






Sheet: /Main FPGA Power/		
File: MainFPGAPower.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.1		Id: 6/20

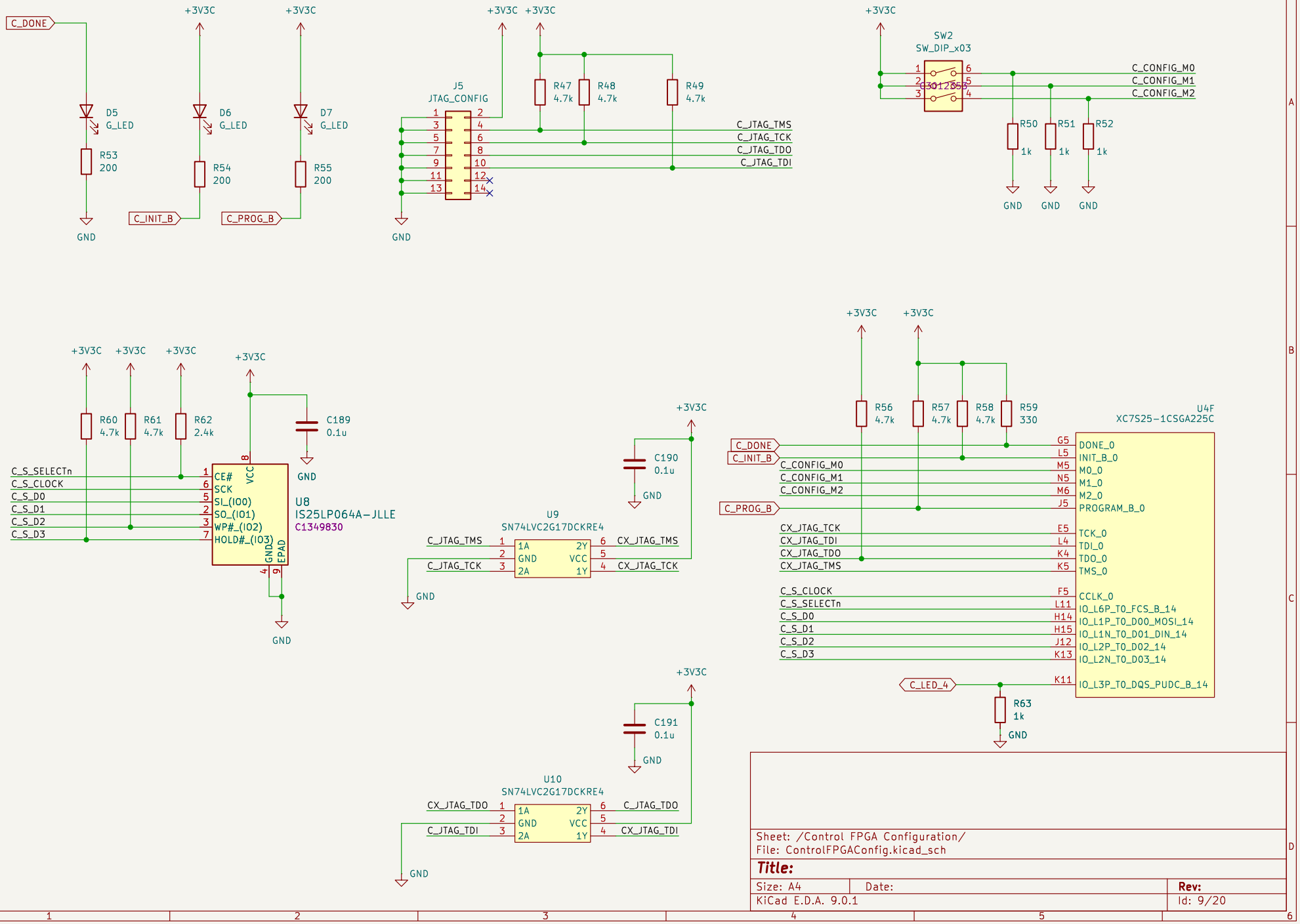

U4A
XC7S25-1CSGA225C
+3V3C
C142 C143 C144 C145 C146 C147 C148
47u 4.7u 4.7u 0.47u 0.47u 0.47u 0.47u
GND
C149 C150 C151 C152 C153 C154 C155
C156 C157 C158 C159 C160 C161 C162
C169 C170 C171 C172 C173 C174 C175
E6 VCCO_0_1
P5 VCCO_0_2
J14 VCCO_14_1
M12 VCCO_14_2
P9 VCCO_14_3
B7 VCCO_15_1
C11 VCCO_15_2
F12 VCCO_15_3
C3 VCCO_34_1
G2 VCCO_34_2
L3 VCCO_34_3
D5 VCCBATT_0
H5 CFGBVS_0
U4G
XC7S25-1CSGA225C
A1 GND_1
A15 GND_2
B8 GND_3
C2 GND_4
C12 GND_5
D6 GND_6
D14 GND_7
E4 GND_8
E7 GND_9
E9 GND_10
F6 GND_11
F10 GND_12
F13 GND_13
G3 GND_14
G7 GND_15
H6 GND_16
GND_17 H10
GND_18 J7
GND_19 J13
GND_20 K6
GND_21 K8
GND_22 K10
GND_23 L2
GND_24 L7
GND_25 L9
GND_26 M11
GND_27 P4
GND_28 P8
GND_29 P13
GND_30 R1
GND_31 R15
U4D
XC7S25-1CSGA225C
+1V8
C163 C164 C165 C166 C167 C168
L6 VCCAUX_1
L8 VCCAUX_2
U4E
XC7S25-1CSGA225C
FB3
MMZ1005S601CT000
C176 0.47u
C177 0.01u
FB4
MMZ1005S601CT000
F9 VCCADC_0
F8 GNDADC_0
G9 VP_0
H8 VN_0
H9 VREFP_0
G8 VREFN_0
U4H
XC7S25-1CSGA225C
J8 DXN_0
J9 DXP_0
U4C
XC7S25-1CSGA225C
+1V0C
C178 C179 C180 C181 C182
4.7u 4.7u 0.47u 0.47u 0.47u
E8 VCCINT_1
E10 VCCINT_2
F7 VCCINT_3
G10 VCCINT_4
H7 VCCINT_5
J10 VCCINT_6
K7 VCCINT_7
K9 VCCINT_8
U4B
XC7S25-1CSGA225C
C183 C184 C185
47u 4.7u 0.47u
J6 VCCBRAM_2
G6 VCCBRAM_1
Sheet: /Controller FPGA Power/
File: ControllerFPGAPower.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.1
Id: 7/20

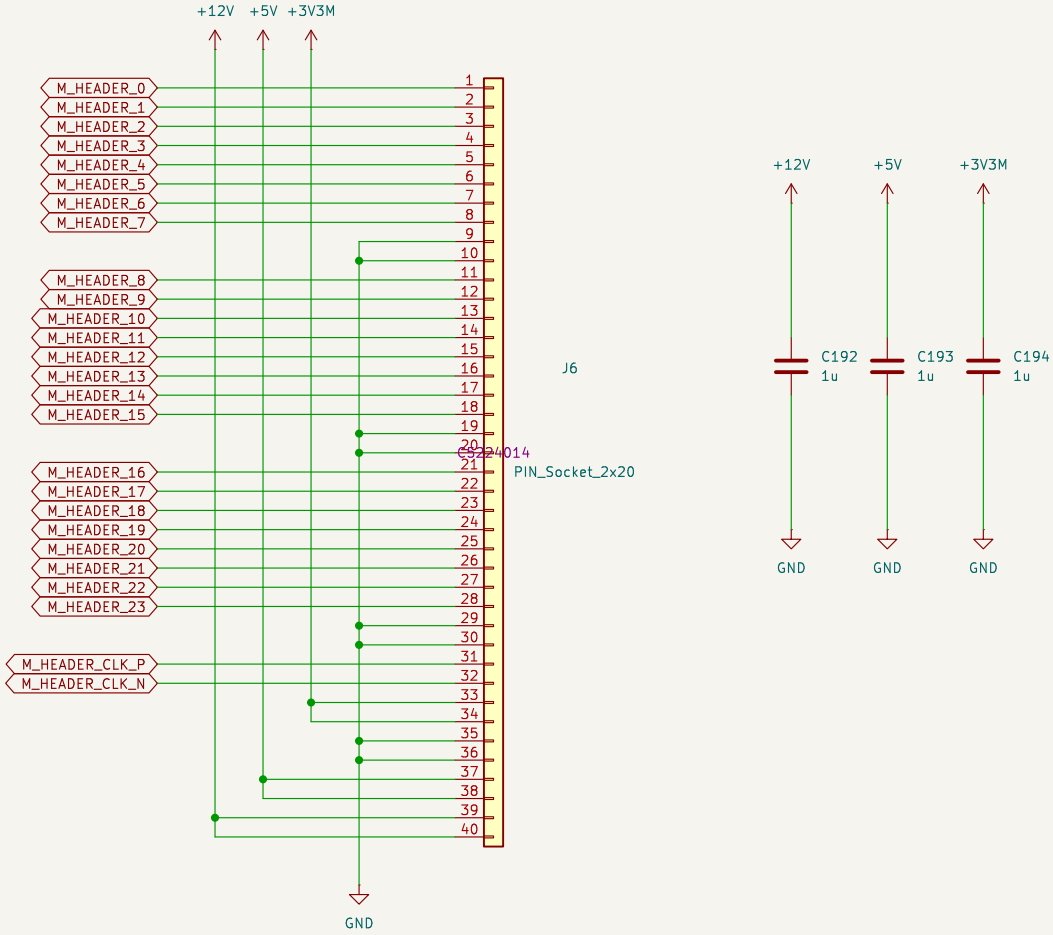


Sheet: /Main FPGA Configuration/
File: MainFPGAConfig.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.1		Id: 8/20


C_DONE
+3V3C
+3V3C
D5
G_LED
D6
G_LED
D7
G_LED
R53
200
R54
200
R55
200
C_INIT_B
C_PROG_B
GND
+3V3C
+3V3C
J5
JTAG_CONFIG
R47
4.7k
R48
4.7k
R49
4.7k
C_JTAG_TMS
C_JTAG_TCK
C_JTAG_TDO
C_JTAG_TDI
GND
+3V3C
SW2
SW_DIP_x03
C_CONFIG_M0
C_CONFIG_M1
C_CONFIG_M2
R50
1k
R51
1k
R52
1k
GND
GND
GND
+3V3C
+3V3C
+3V3C
+3V3C
R60
4.7k
R61
4.7k
R62
2.4k
C189
0.1u
C_S_SELECTn
C_S_CLOCK
C_S_D0
C_S_D1
C_S_D2
C_S_D3
CE#
SCK
SI_(IO0)
SO_(IO1)
WP#_(IO2)
HOLD#_(IO3)
VCC
GND
EPAD
U8
IS25LP064A-JLLE
C1349830
GND
GND
+3V3C
C190
0.1u
GND
U9
SN74LVC2G17DCKRE4
C_JTAG_TMS
C_JTAG_TCK
1A
GND
2A
2Y
VCC
1Y
CX_JTAG_TMS
CX_JTAG_TCK
GND
+3V3C
C191
0.1u
GND
U10
SN74LVC2G17DCKRE4
CX_JTAG_TDO
C_JTAG_TDI
1A
GND
2A
2Y
VCC
1Y
C_JTAG_TDO
CX_JTAG_TDI
GND
+3V3C
+3V3C
R56
4.7k
R57
4.7k
R58
4.7k
R59
330
U4F
XC7S25-1CSGA225C
C_DONE
C_INIT_B
C_CONFIG_M0
C_CONFIG_M1
C_CONFIG_M2
C_PROG_B
CX_JTAG_TCK
CX_JTAG_TDI
CX_JTAG_TDO
CX_JTAG_TMS
C_S_CLOCK
C_S_SELECTn
C_S_D0
C_S_D1
C_S_D2
C_S_D3
C_LED_4
G5
L5
M5
N5
M6
J5
E5
L4
K4
K5
F5
L11
H14
H15
J12
K13
K11
DONE_0
INIT_B_0
M0_0
M1_0
M2_0
PROGRAM_B_0
TCK_0
TDI_0
TDO_0
TMS_0
CCLK_0
IO_L6P_T0_FCS_B_14
IO_L1P_T0_D00_MOSI_14
IO_L1N_T0_D01_DIN_14
IO_L2P_T0_D02_14
IO_L2N_T0_D03_14
IO_L3P_T0_DQS_PUDC_B_14
R63
1k
GND
Sheet: /Control FPGA Configuration/
File: ControlFPGAConfig.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.1
Id: 9/20

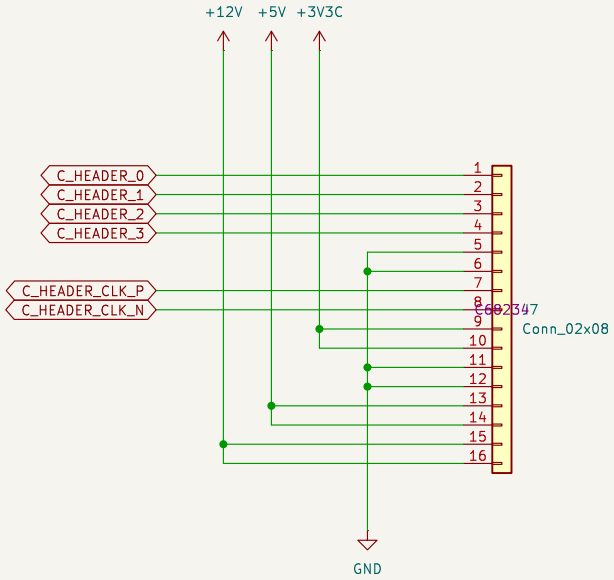


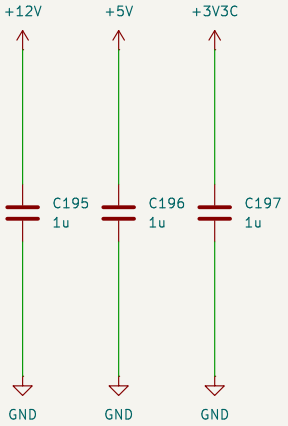
Sheet: /Main FPGA Header/
File: MainFPGAHeader.kicad_sch

Title:

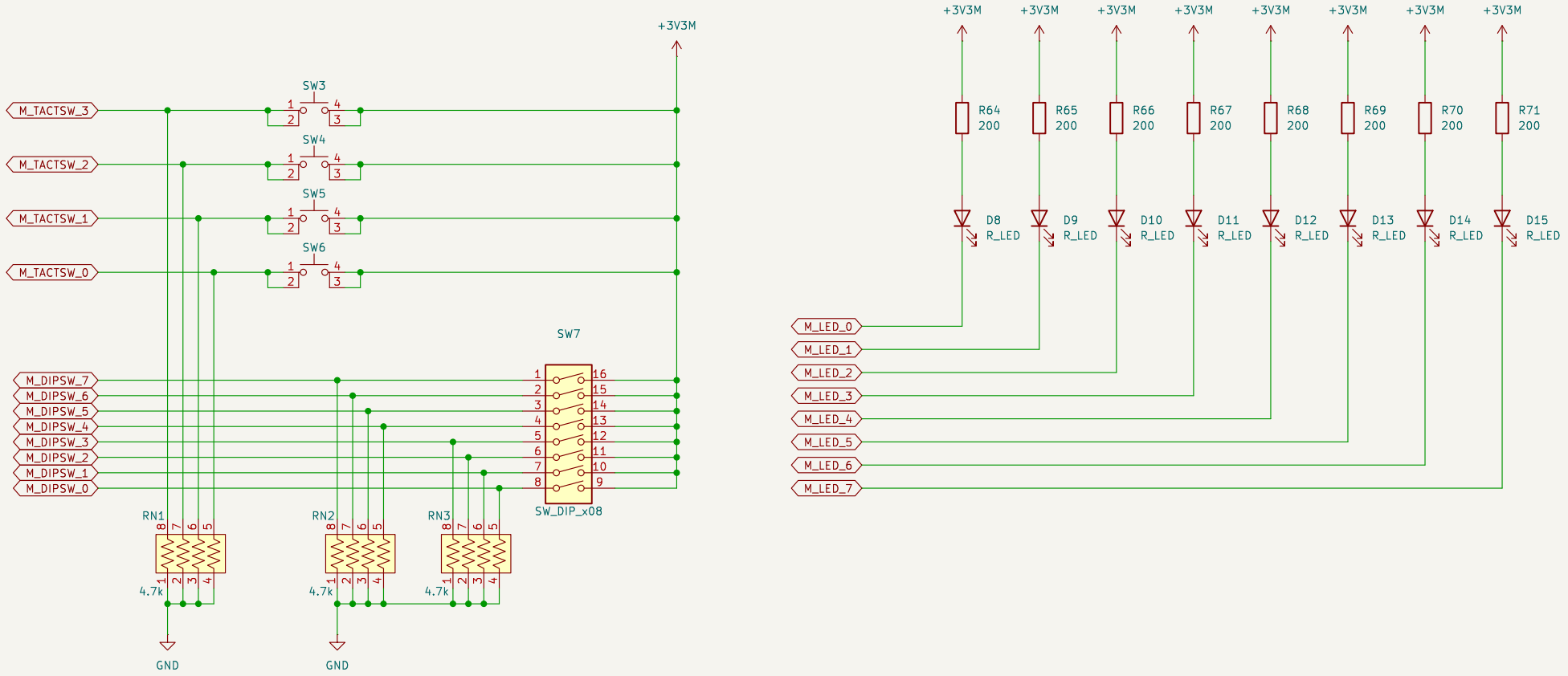
Size: A4 | Date: | **Rev:**

KiCad E.D.A. 9.0.1 | Id: 9/20





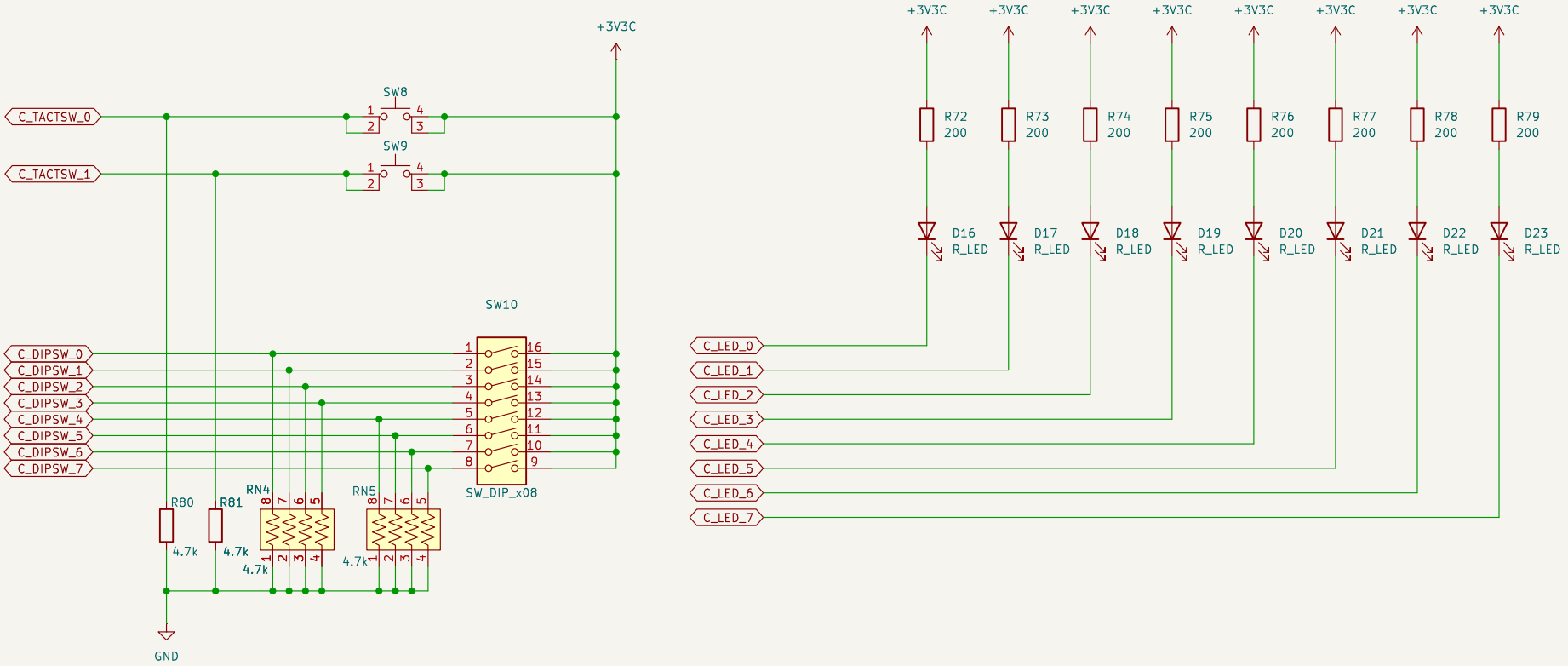
Sheet: /Controller FPGA Header/ File: ControllerFPGAHeader.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.1		Id: 11/20



Sheet: /Main FPGA Peripheral/
File: MainFPGAPeripheral.kicad_sch

Title:

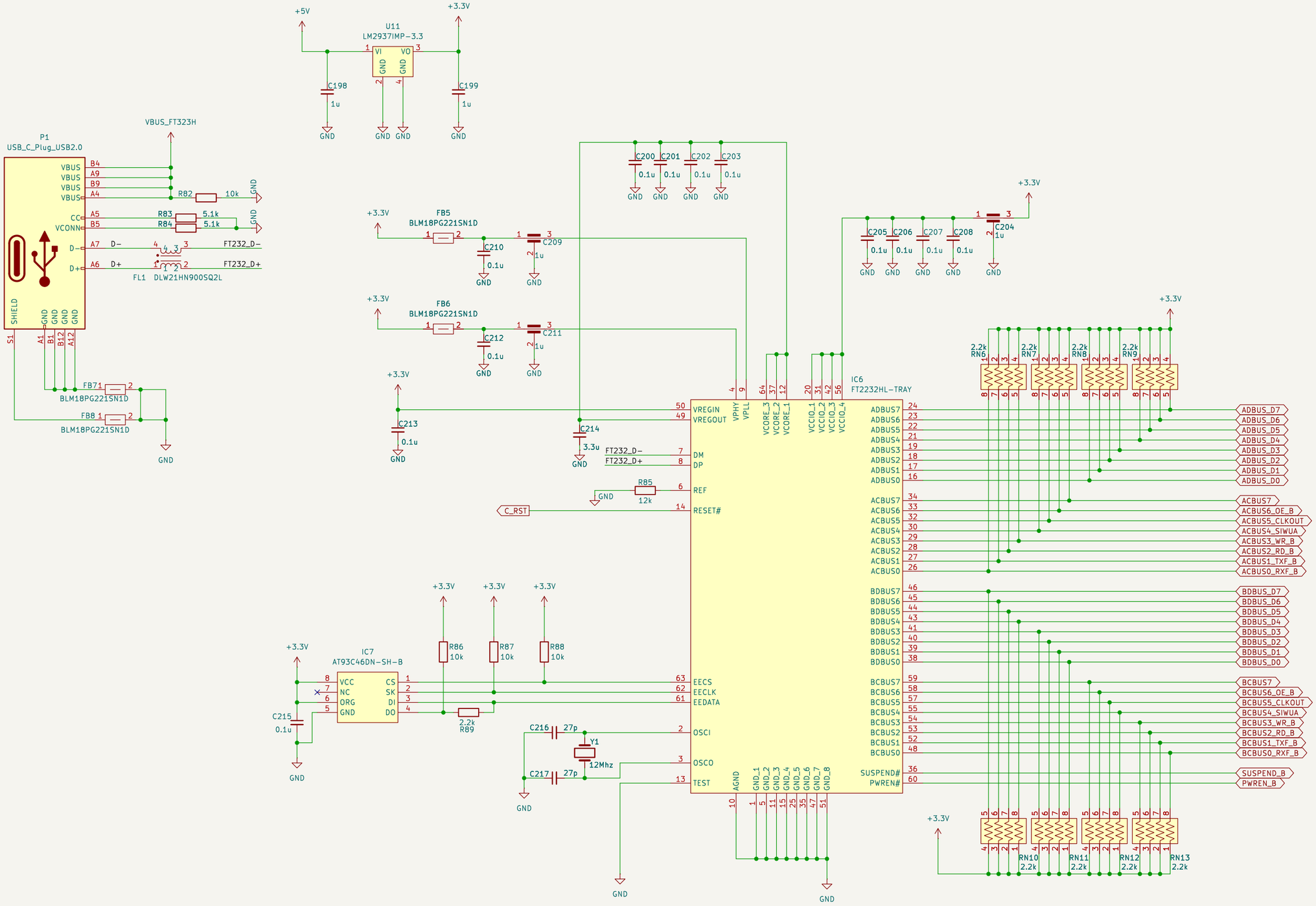
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.1		Id: 12/20



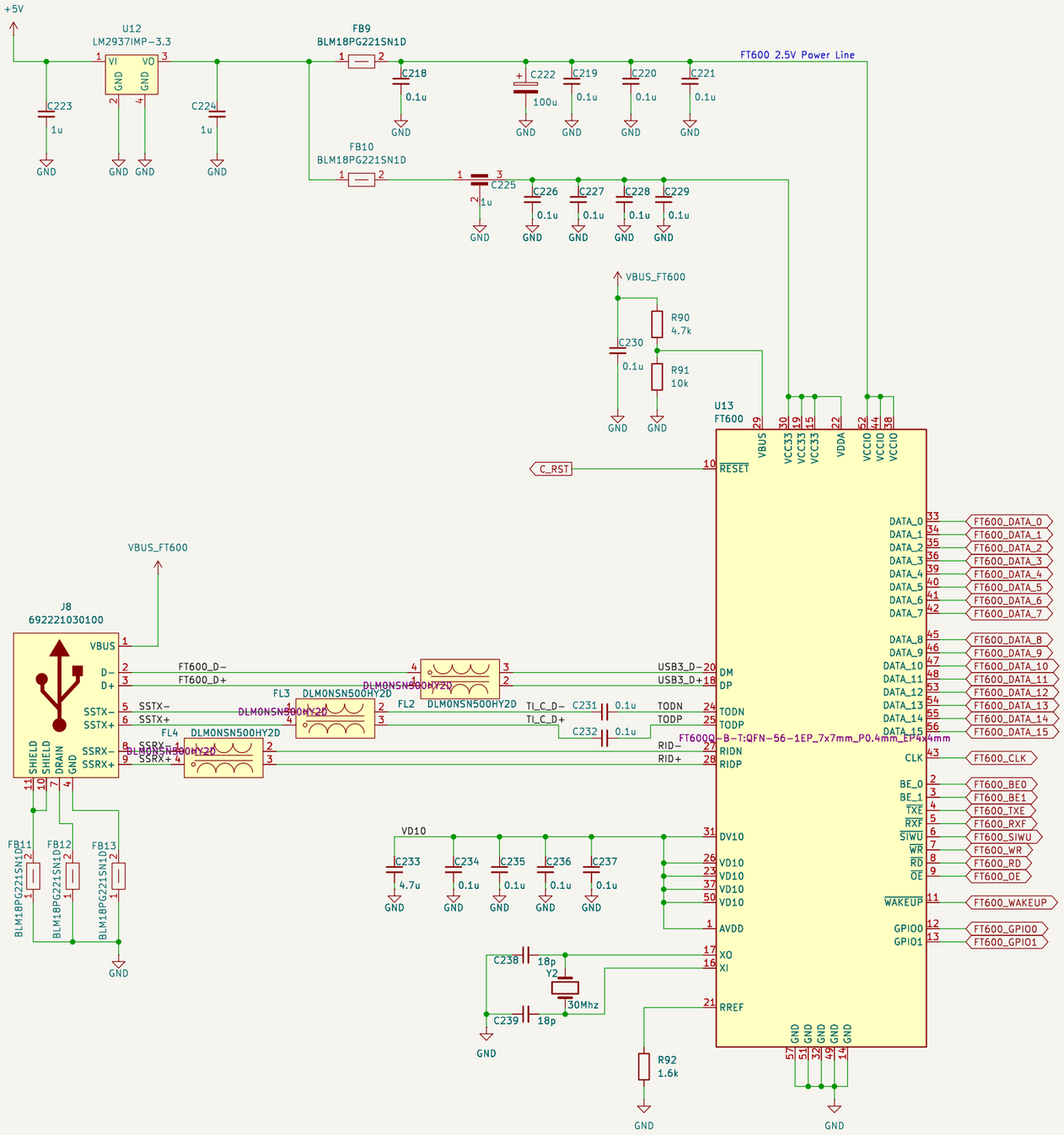
Sheet: /Control FPGA Peripheral/
File: ControlFPGAPeripheral.kicad_sch

Title:

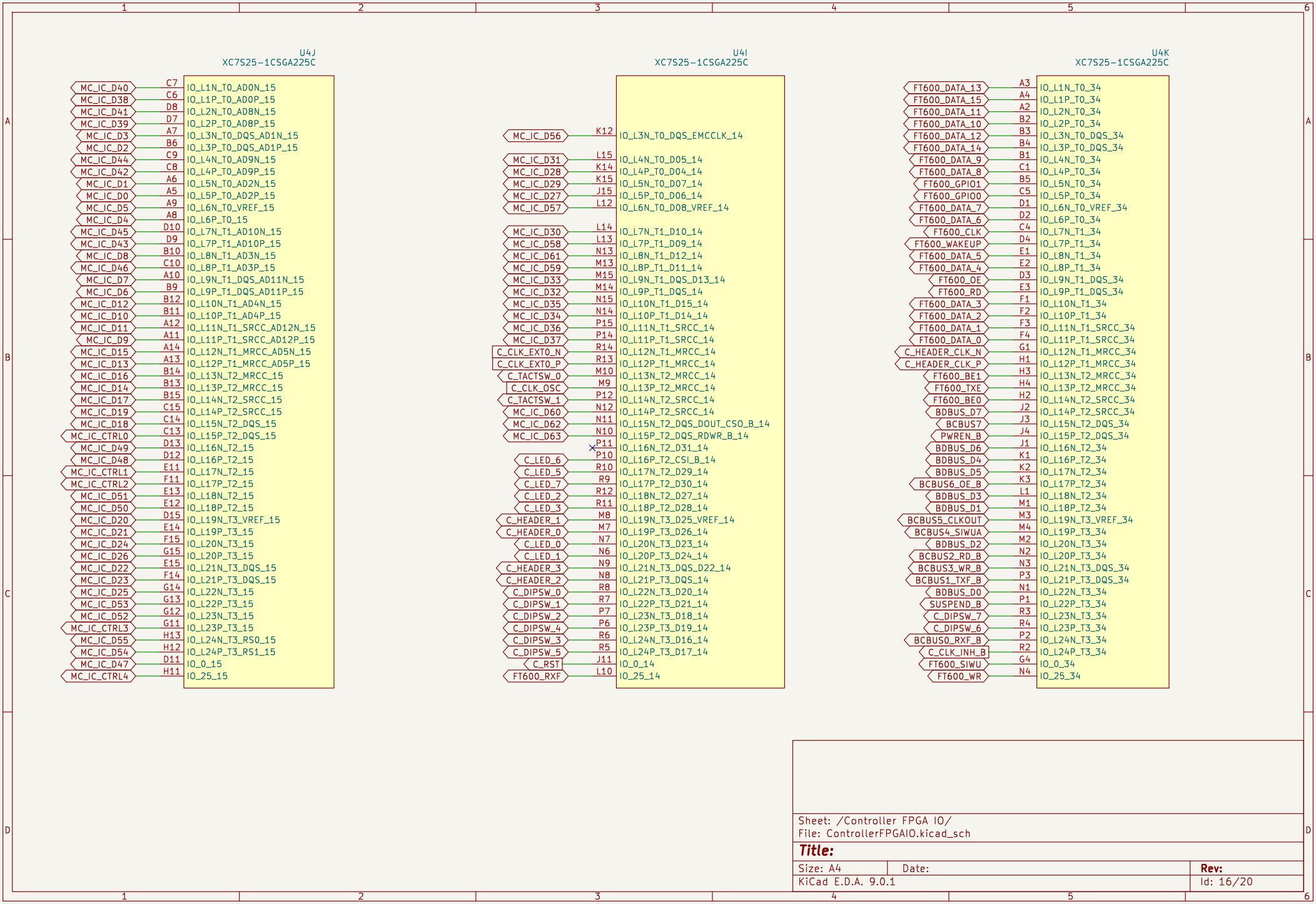
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.1		Id: 13/20

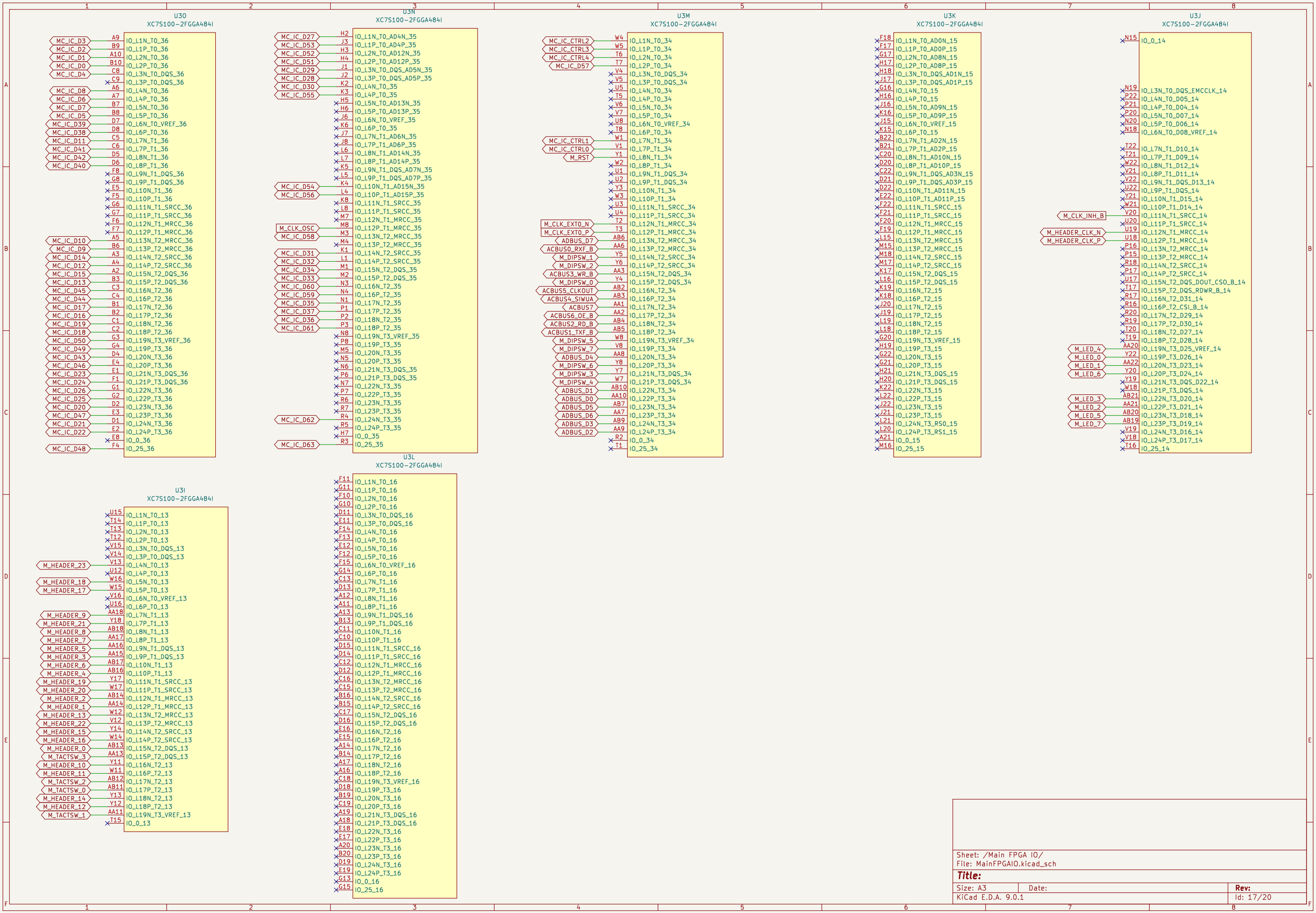


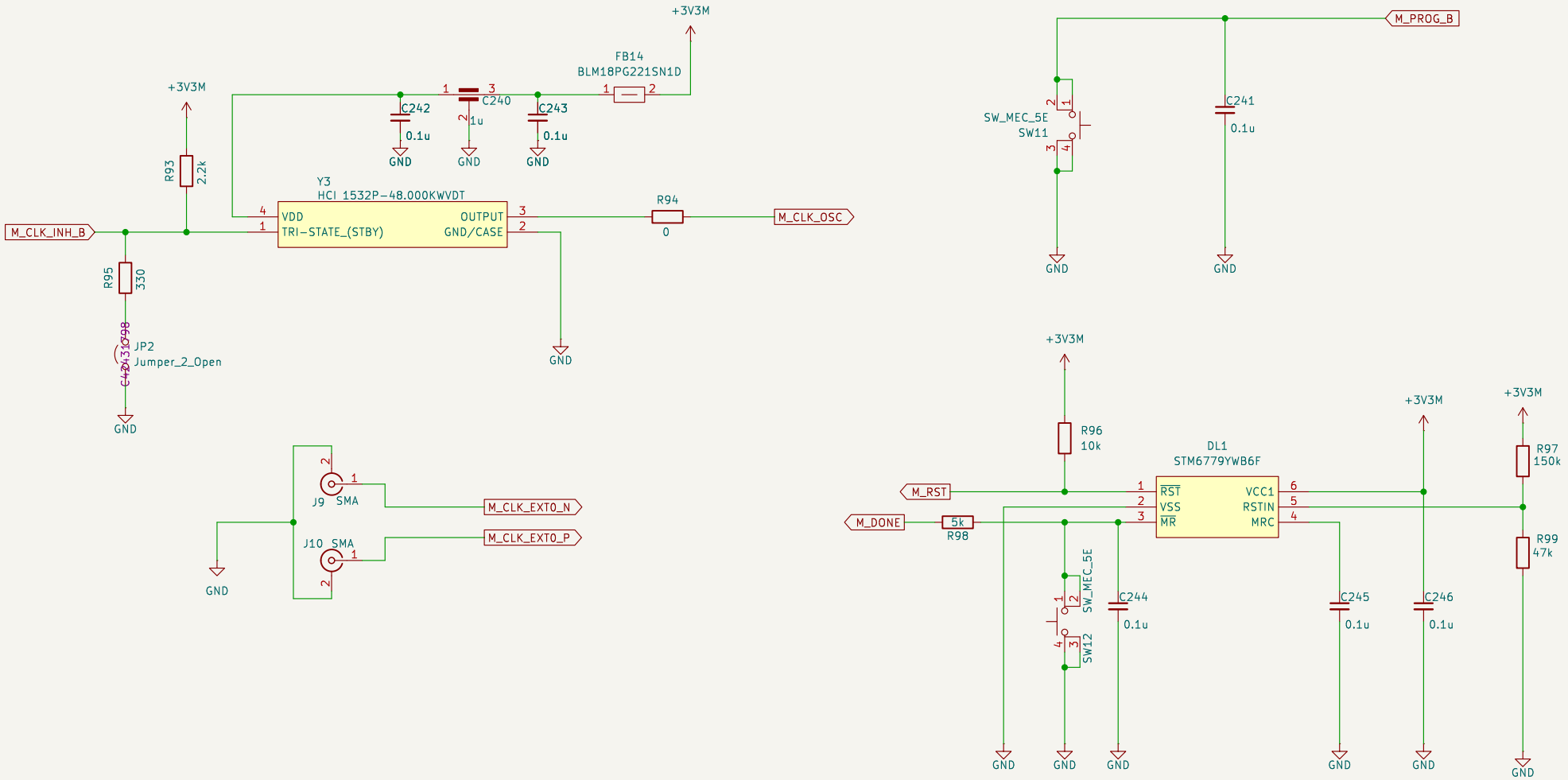
Sheet: /FT2232HL/		
File: FT2232HL.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.1		Id: 14/20



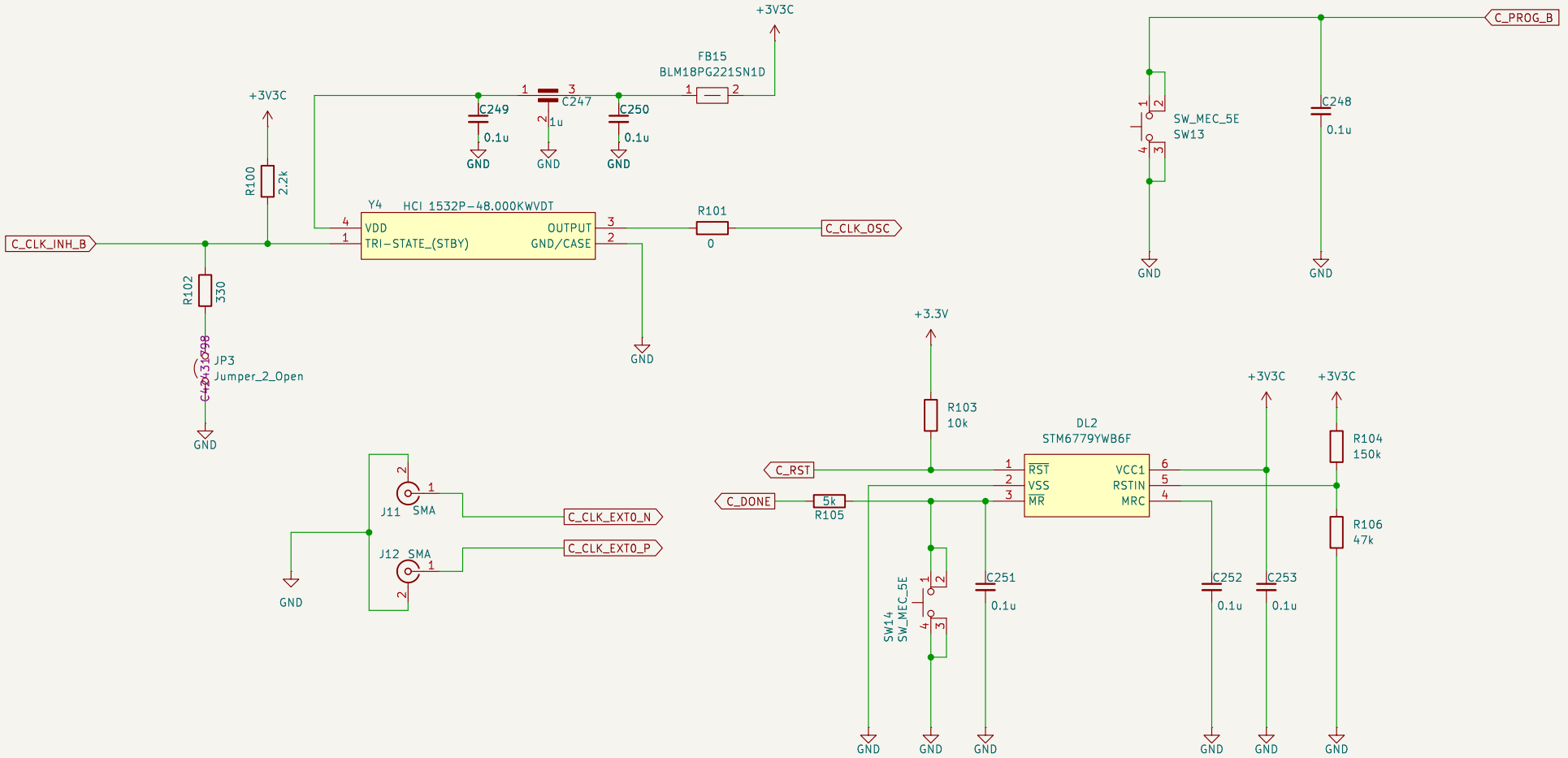
Sheet: /FT600Q/		
File: FT600Q.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.1		Id: 15/20


U4J
XC7S25-1CSGA225C
MC_IC_D40 C7 IO_L1N_T0_AD0N_15
MC_IC_D38 C6 IO_L1P_T0_AD0P_15
MC_IC_D41 D8 IO_L2N_T0_AD8N_15
MC_IC_D39 D7 IO_L2P_T0_AD8P_15
MC_IC_D3 A7 IO_L3N_T0_DQS_AD1N_15
MC_IC_D2 B6 IO_L3P_T0_DQS_AD1P_15
MC_IC_D44 C9 IO_L4N_T0_AD9N_15
MC_IC_D42 C8 IO_L4P_T0_AD9P_15
MC_IC_D1 A6 IO_L5N_T0_AD2N_15
MC_IC_D0 A5 IO_L5P_T0_AD2P_15
MC_IC_D5 A9 IO_L6N_T0_VREF_15
MC_IC_D4 A8 IO_L6P_T0_15
MC_IC_D45 D10 IO_L7N_T1_AD10N_15
MC_IC_D43 D9 IO_L7P_T1_AD10P_15
MC_IC_D8 B10 IO_L8N_T1_AD3N_15
MC_IC_D46 C10 IO_L8P_T1_AD3P_15
MC_IC_D7 A10 IO_L9N_T1_DQS_AD11N_15
MC_IC_D6 B9 IO_L9P_T1_DQS_AD11P_15
MC_IC_D12 B12 IO_L10N_T1_AD4N_15
MC_IC_D10 B11 IO_L10P_T1_AD4P_15
MC_IC_D11 A12 IO_L11N_T1_SRCC_AD12N_15
MC_IC_D9 A11 IO_L11P_T1_SRCC_AD12P_15
MC_IC_D15 A14 IO_L12N_T1_MRCC_AD5N_15
MC_IC_D13 A13 IO_L12P_T1_MRCC_AD5P_15
MC_IC_D16 B14 IO_L13N_T2_MRCC_15
MC_IC_D14 B13 IO_L13P_T2_MRCC_15
MC_IC_D17 B15 IO_L14N_T2_SRCC_15
MC_IC_D19 C15 IO_L14P_T2_SRCC_15
MC_IC_D18 C14 IO_L15N_T2_DQS_15
MC_IC_CTRL0 C13 IO_L15P_T2_DQS_15
MC_IC_D49 D13 IO_L16N_T2_15
MC_IC_D48 D12 IO_L16P_T2_15
MC_IC_CTRL1 E11 IO_L17N_T2_15
MC_IC_CTRL2 F11 IO_L17P_T2_15
MC_IC_D51 E13 IO_L18N_T2_15
MC_IC_D50 E12 IO_L18P_T2_15
MC_IC_D20 D15 IO_L19N_T3_VREF_15
MC_IC_D21 E14 IO_L19P_T3_15
MC_IC_D24 F15 IO_L20N_T3_15
MC_IC_D26 G15 IO_L20P_T3_15
MC_IC_D22 E15 IO_L21N_T3_DQS_15
MC_IC_D23 F14 IO_L21P_T3_DQS_15
MC_IC_D25 G14 IO_L22N_T3_15
MC_IC_D53 G13 IO_L22P_T3_15
MC_IC_D52 G12 IO_L23N_T3_15
MC_IC_CTRL3 G11 IO_L23P_T3_15
MC_IC_D55 H13 IO_L24N_T3_RS0_15
MC_IC_D54 H12 IO_L24P_T3_RS1_15
MC_IC_D47 D11 IO_0_15
MC_IC_CTRL4 H11 IO_25_15
U4I
XC7S25-1CSGA225C
MC_IC_D56 K12 IO_L3N_T0_DQS_EMCCLK_14
MC_IC_D31 L15 IO_L4N_T0_D05_14
MC_IC_D28 K14 IO_L4P_T0_D04_14
MC_IC_D29 K15 IO_L5N_T0_D07_14
MC_IC_D27 J15 IO_L5P_T0_D06_14
MC_IC_D57 L12 IO_L6N_T0_D08_VREF_14
MC_IC_D30 L14 IO_L7N_T1_D10_14
MC_IC_D58 L13 IO_L7P_T1_D09_14
MC_IC_D61 N13 IO_L8N_T1_D12_14
MC_IC_D59 M13 IO_L8P_T1_D11_14
MC_IC_D33 M15 IO_L9N_T1_DQS_D13_14
MC_IC_D32 M14 IO_L9P_T1_DQS_14
MC_IC_D35 N15 IO_L10N_T1_D15_14
MC_IC_D34 N14 IO_L10P_T1_D14_14
MC_IC_D36 P15 IO_L11N_T1_SRCC_14
MC_IC_D37 P14 IO_L11P_T1_SRCC_14
C_CLK_EXT0_N R14 IO_L12N_T1_MRCC_14
C_CLK_EXT0_P R13 IO_L12P_T1_MRCC_14
C_TACTSW_0 M10 IO_L13N_T2_MRCC_14
C_CLK_OSC M9 IO_L13P_T2_MRCC_14
C_TACTSW_1 P12 IO_L14N_T2_SRCC_14
MC_IC_D60 N12 IO_L14P_T2_SRCC_14
MC_IC_D62 N11 IO_L15N_T2_DQS_DOUT_CSO_B_14
MC_IC_D63 N10 IO_L15P_T2_DQS_RDWR_B_14
P11 IO_L16N_T2_D31_14
C_LED_6 P10 IO_L16P_T2_CSI_B_14
C_LED_5 R10 IO_L17N_T2_D29_14
C_LED_7 R9 IO_L17P_T2_D30_14
C_LED_2 R12 IO_L18N_T2_D27_14
C_LED_3 R11 IO_L18P_T2_D28_14
C_HEADER_1 M8 IO_L19N_T3_D25_VREF_14
C_HEADER_0 M7 IO_L19P_T3_D26_14
C_LED_0 N7 IO_L20N_T3_D23_14
C_LED_1 N6 IO_L20P_T3_D24_14
C_HEADER_3 N9 IO_L21N_T3_DQS_D22_14
C_HEADER_2 N8 IO_L21P_T3_DQS_14
C_DIPSW_0 R8 IO_L22N_T3_D20_14
C_DIPSW_1 R7 IO_L22P_T3_D21_14
C_DIPSW_2 P7 IO_L23N_T3_D18_14
C_DIPSW_4 P6 IO_L23P_T3_D19_14
C_DIPSW_3 R6 IO_L24N_T3_D16_14
C_DIPSW_5 R5 IO_L24P_T3_D17_14
C_RST J11 IO_0_14
FT600_RXF L10 IO_25_14
U4K
XC7S25-1CSGA225C
FT600_DATA_13 A3 IO_L1N_T0_34
FT600_DATA_15 A4 IO_L1P_T0_34
FT600_DATA_11 A2 IO_L2N_T0_34
FT600_DATA_10 B2 IO_L2P_T0_34
FT600_DATA_12 B3 IO_L3N_T0_DQS_34
FT600_DATA_14 B4 IO_L3P_T0_DQS_34
FT600_DATA_9 B1 IO_L4N_T0_34
FT600_DATA_8 C1 IO_L4P_T0_34
FT600_GPIO1 B5 IO_L5N_T0_34
FT600_GPIO0 C5 IO_L5P_T0_34
FT600_DATA_7 D1 IO_L6N_T0_VREF_34
FT600_DATA_6 D2 IO_L6P_T0_34
FT600_CLK C4 IO_L7N_T1_34
FT600_WAKEUP D4 IO_L7P_T1_34
FT600_DATA_5 E1 IO_L8N_T1_34
FT600_DATA_4 E2 IO_L8P_T1_34
FT600_OE D3 IO_L9N_T1_DQS_34
FT600_RD E3 IO_L9P_T1_DQS_34
FT600_DATA_3 F1 IO_L10N_T1_34
FT600_DATA_2 F2 IO_L10P_T1_34
FT600_DATA_1 F3 IO_L11N_T1_SRCC_34
FT600_DATA_0 F4 IO_L11P_T1_SRCC_34
C_HEADER_CLK_N G1 IO_L12N_T1_MRCC_34
C_HEADER_CLK_P H1 IO_L12P_T1_MRCC_34
FT600_BE1 H3 IO_L13N_T2_MRCC_34
FT600_TXE H4 IO_L13P_T2_MRCC_34
FT600_BE0 H2 IO_L14N_T2_SRCC_34
BDBUS_D7 J2 IO_L14P_T2_SRCC_34
BCBUS7 J3 IO_L15N_T2_DQS_34
PWREN_B J4 IO_L15P_T2_DQS_34
BDBUS_D6 J1 IO_L16N_T2_34
BDBUS_D4 K1 IO_L16P_T2_34
BDBUS_D5 K2 IO_L17N_T2_34
BCBUS6_OE_B K3 IO_L17P_T2_34
BDBUS_D3 L1 IO_L18N_T2_34
BDBUS_D1 M1 IO_L18P_T2_34
BCBUS5_CLKOUT M3 IO_L19N_T3_VREF_34
BCBUS4_SIWUA M4 IO_L19P_T3_34
BDBUS_D2 M2 IO_L20N_T3_34
BCBUS2_RD_B N2 IO_L20P_T3_34
BCBUS3_WR_B N3 IO_L21N_T3_DQS_34
BCBUS1_TXF_B P3 IO_L21P_T3_DQS_34
BDBUS_D0 N1 IO_L22N_T3_34
SUSPEND_B P1 IO_L22P_T3_34
C_DIPSW_7 R3 IO_L23N_T3_34
C_DIPSW_6 R4 IO_L23P_T3_34
BCBUS0_RXF_B P2 IO_L24N_T3_34
C_CLK_INH_B R2 IO_L24P_T3_34
FT600_SIWU G4 IO_0_34
FT600_WR N4 IO_25_34
Sheet: /Controller FPGA IO/
File: ControllerFPGAIO.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.1
Id: 16/20

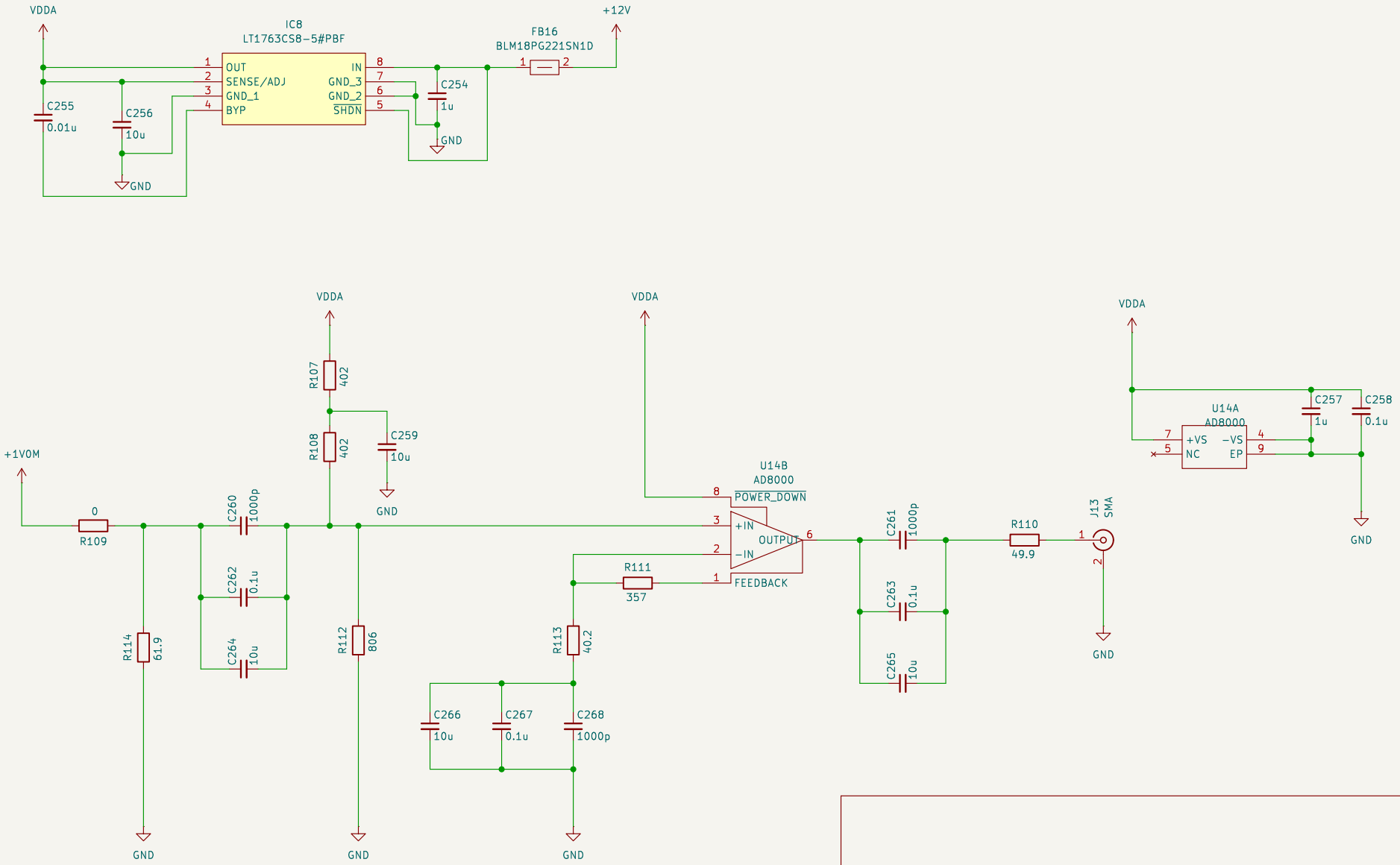

U3O
XC7S100-2FGGA484I
MC_IC_D3 A9 IO_L1N_T0_36
MC_IC_D2 B9 IO_L1P_T0_36
MC_IC_D1 A10 IO_L2N_T0_36
MC_IC_D0 B10 IO_L2P_T0_36
MC_IC_D4 C8 IO_L3N_T0_DQS_36
C9 IO_L3P_T0_DQS_36
MC_IC_D8 A6 IO_L4N_T0_36
MC_IC_D6 A7 IO_L4P_T0_36
MC_IC_D7 B7 IO_L5N_T0_36
MC_IC_D5 B8 IO_L5P_T0_36
MC_IC_D39 D7 IO_L6N_T0_VREF_36
MC_IC_D38 D8 IO_L6P_T0_36
MC_IC_D11 C5 IO_L7N_T1_36
MC_IC_D41 C6 IO_L7P_T1_36
MC_IC_D42 D5 IO_L8N_T1_36
MC_IC_D40 D6 IO_L8P_T1_36
F8 IO_L9N_T1_DQS_36
G8 IO_L9P_T1_DQS_36
E5 IO_L10N_T1_36
F5 IO_L10P_T1_36
G6 IO_L11N_T1_SRCC_36
G7 IO_L11P_T1_SRCC_36
F6 IO_L12N_T1_MRCC_36
F7 IO_L12P_T1_MRCC_36
MC_IC_D10 A5 IO_L13N_T2_MRCC_36
MC_IC_D9 B6 IO_L13P_T2_MRCC_36
MC_IC_D14 A3 IO_L14N_T2_SRCC_36
MC_IC_D12 A4 IO_L14P_T2_SRCC_36
MC_IC_D15 A2 IO_L15N_T2_DQS_36
MC_IC_D13 B3 IO_L15P_T2_DQS_36
MC_IC_D45 C3 IO_L16N_T2_36
MC_IC_D44 C4 IO_L16P_T2_36
MC_IC_D17 B1 IO_L17N_T2_36
MC_IC_D16 B2 IO_L17P_T2_36
MC_IC_D19 C1 IO_L18N_T2_36
MC_IC_D18 C2 IO_L18P_T2_36
MC_IC_D50 G3 IO_L19N_T3_VREF_36
MC_IC_D49 G4 IO_L19P_T3_36
MC_IC_D43 D4 IO_L20N_T3_36
MC_IC_D46 E4 IO_L20P_T3_36
MC_IC_D23 E1 IO_L21N_T3_DQS_36
MC_IC_D24 F1 IO_L21P_T3_DQS_36
MC_IC_D26 G1 IO_L22N_T3_36
MC_IC_D25 G2 IO_L22P_T3_36
MC_IC_D20 D2 IO_L23N_T3_36
MC_IC_D47 E3 IO_L23P_T3_36
MC_IC_D21 D1 IO_L24N_T3_36
MC_IC_D22 E2 IO_L24P_T3_36
E8 IO_0_36
MC_IC_D48 F4 IO_25_36
U3N
XC7S100-2FGGA484I
MC_IC_D27 H2 IO_L1N_T0_AD4N_35
MC_IC_D53 J3 IO_L1P_T0_AD4P_35
MC_IC_D52 H3 IO_L2N_T0_AD12N_35
MC_IC_D51 H4 IO_L2P_T0_AD12P_35
MC_IC_D29 J1 IO_L3N_T0_DQS_AD5N_35
MC_IC_D28 J2 IO_L3P_T0_DQS_AD5P_35
MC_IC_D30 K2 IO_L4N_T0_35
MC_IC_D55 K3 IO_L4P_T0_35
H5 IO_L5N_T0_AD13N_35
H6 IO_L5P_T0_AD13P_35
J6 IO_L6N_T0_VREF_35
K6 IO_L6P_T0_35
J7 IO_L7N_T1_AD6N_35
J8 IO_L7P_T1_AD6P_35
L6 IO_L8N_T1_AD14N_35
L7 IO_L8P_T1_AD14P_35
K5 IO_L9N_T1_DQS_AD7N_35
L5 IO_L9P_T1_DQS_AD7P_35
MC_IC_D54 K4 IO_L10N_T1_AD15N_35
MC_IC_D56 L4 IO_L10P_T1_AD15P_35
K8 IO_L11N_T1_SRCC_35
L8 IO_L11P_T1_SRCC_35
M7 IO_L12N_T1_MRCC_35
M8 IO_L12P_T1_MRCC_35
M_CLK_OSC M3 IO_L13N_T2_MRCC_35
MC_IC_D58 M4 IO_L13P_T2_MRCC_35
MC_IC_D31 K1 IO_L14N_T2_SRCC_35
MC_IC_D32 L1 IO_L14P_T2_SRCC_35
MC_IC_D34 M1 IO_L15N_T2_DQS_35
MC_IC_D33 M2 IO_L15P_T2_DQS_35
MC_IC_D60 N3 IO_L16N_T2_35
MC_IC_D59 N4 IO_L16P_T2_35
MC_IC_D35 N1 IO_L17N_T2_35
MC_IC_D37 P1 IO_L17P_T2_35
MC_IC_D36 P2 IO_L18N_T2_35
MC_IC_D61 P3 IO_L18P_T2_35
N8 IO_L19N_T3_VREF_35
P8 IO_L19P_T3_35
M5 IO_L20N_T3_35
N5 IO_L20P_T3_35
N6 IO_L21N_T3_DQS_35
P6 IO_L21P_T3_DQS_35
N7 IO_L22N_T3_35
P7 IO_L22P_T3_35
R6 IO_L23N_T3_35
R7 IO_L23P_T3_35
MC_IC_D62 R4 IO_L24N_T3_35
R5 IO_L24P_T3_35
H7 IO_0_35
MC_IC_D63 R3 IO_25_35
U3M
XC7S100-2FGGA484I
MC_IC_CTRL2 W4 IO_L1N_T0_34
MC_IC_CTRL3 W5 IO_L1P_T0_34
MC_IC_CTRL4 T6 IO_L2N_T0_34
MC_IC_D57 T7 IO_L2P_T0_34
V4 IO_L3N_T0_DQS_34
V5 IO_L3P_T0_DQS_34
U5 IO_L4N_T0_34
T5 IO_L4P_T0_34
V6 IO_L5N_T0_34
V7 IO_L5P_T0_34
U8 IO_L6N_T0_VREF_34
T8 IO_L6P_T0_34
MC_IC_CTRL1 W1 IO_L7N_T1_34
MC_IC_CTRL0 V1 IO_L7P_T1_34
M_RST Y1 IO_L8N_T1_34
W2 IO_L8P_T1_34
U1 IO_L9N_T1_DQS_34
U2 IO_L9P_T1_DQS_34
Y3 IO_L10N_T1_34
W3 IO_L10P_T1_34
U3 IO_L11N_T1_SRCC_34
U4 IO_L11P_T1_SRCC_34
M_CLK_EXT0_N T2 IO_L12N_T1_MRCC_34
M_CLK_EXT0_P T3 IO_L12P_T1_MRCC_34
ADBUS_D7 AB6 IO_L13N_T2_MRCC_34
ACBUS0_RXF_B AA6 IO_L13P_T2_MRCC_34
M_DIPSW_1 Y5 IO_L14N_T2_SRCC_34
M_DIPSW_2 Y6 IO_L14P_T2_SRCC_34
ACBUS3_WR_B AA3 IO_L15N_T2_DQS_34
M_DIPSW_0 Y4 IO_L15P_T2_DQS_34
ACBUS5_CLKOUT AB2 IO_L16N_T2_34
ACBUS4_SIWUA AB3 IO_L16P_T2_34
ACBUS7 AA1 IO_L17N_T2_34
ACBUS6_OE_B AA2 IO_L17P_T2_34
ACBUS2_RD_B AB4 IO_L18N_T2_34
ACBUS1_TXF_B AB5 IO_L18P_T2_34
M_DIPSW_5 W8 IO_L19N_T3_VREF_34
M_DIPSW_7 V8 IO_L19P_T3_34
ADBUS_D4 AA8 IO_L20N_T3_34
M_DIPSW_6 Y8 IO_L20P_T3_34
M_DIPSW_3 Y7 IO_L21N_T3_DQS_34
M_DIPSW_4 W7 IO_L21P_T3_DQS_34
ADBUS_D1 AB10 IO_L22N_T3_34
ADBUS_D0 AA10 IO_L22P_T3_34
ADBUS_D5 AB7 IO_L23N_T3_34
ADBUS_D6 AA7 IO_L23P_T3_34
ADBUS_D3 AB9 IO_L24N_T3_34
ADBUS_D2 AA9 IO_L24P_T3_34
R2 IO_0_34
T1 IO_25_34
U3K
XC7S100-2FGGA484I
F18 IO_L1N_T0_AD0N_15
F17 IO_L1P_T0_AD0P_15
G17 IO_L2N_T0_AD8N_15
H17 IO_L2P_T0_AD8P_15
H18 IO_L3N_T0_DQS_AD1N_15
J17 IO_L3P_T0_DQS_AD1P_15
G16 IO_L4N_T0_15
H16 IO_L4P_T0_15
J16 IO_L5N_T0_AD9N_15
K16 IO_L5P_T0_AD9P_15
J15 IO_L6N_T0_VREF_15
K15 IO_L6P_T0_15
B22 IO_L7N_T1_AD2N_15
B21 IO_L7P_T1_AD2P_15
C20 IO_L8N_T1_AD10N_15
D20 IO_L8P_T1_AD10P_15
C22 IO_L9N_T1_DQS_AD3N_15
D21 IO_L9P_T1_DQS_AD3P_15
D22 IO_L10N_T1_AD11N_15
E22 IO_L10P_T1_AD11P_15
F22 IO_L11N_T1_SRCC_15
F21 IO_L11P_T1_SRCC_15
F20 IO_L12N_T1_MRCC_15
F19 IO_L12P_T1_MRCC_15
L15 IO_L13N_T2_MRCC_15
M15 IO_L13P_T2_MRCC_15
M18 IO_L14N_T2_SRCC_15
M17 IO_L14P_T2_SRCC_15
K17 IO_L15N_T2_DQS_15
L16 IO_L15P_T2_DQS_15
K19 IO_L16N_T2_15
K18 IO_L16P_T2_15
J20 IO_L17N_T2_15
J19 IO_L17P_T2_15
L19 IO_L18N_T2_15
L18 IO_L18P_T2_15
G20 IO_L19N_T3_VREF_15
H19 IO_L19P_T3_15
G22 IO_L20N_T3_15
G21 IO_L20P_T3_15
H21 IO_L21N_T3_DQS_15
H20 IO_L21P_T3_DQS_15
K22 IO_L22N_T3_15
L22 IO_L22P_T3_15
J22 IO_L23N_T3_15
J21 IO_L23P_T3_15
L21 IO_L24N_T3_RS0_15
L20 IO_L24P_T3_RS1_15
A21 IO_0_15
M16 IO_25_15
U3J
XC7S100-2FGGA484I
N15 IO_0_14
N19 IO_L3N_T0_DQS_EMCCLK_14
P22 IO_L4N_T0_D05_14
P21 IO_L4P_T0_D04_14
P20 IO_L5N_T0_D07_14
N20 IO_L5P_T0_D06_14
N18 IO_L6N_T0_D08_VREF_14
T22 IO_L7N_T1_D10_14
T21 IO_L7P_T1_D09_14
W22 IO_L8N_T1_D12_14
V21 IO_L8P_T1_D11_14
V22 IO_L9N_T1_DQS_D13_14
U22 IO_L9P_T1_DQS_14
Y21 IO_L10N_T1_D15_14
W21 IO_L10P_T1_D14_14
M_CLK_INH_B V20 IO_L11N_T1_SRCC_14
U20 IO_L11P_T1_SRCC_14
M_HEADER_CLK_N U19 IO_L12N_T1_MRCC_14
M_HEADER_CLK_P U18 IO_L12P_T1_MRCC_14
P16 IO_L13N_T2_MRCC_14
P15 IO_L13P_T2_MRCC_14
R18 IO_L14N_T2_SRCC_14
P17 IO_L14P_T2_SRCC_14
U17 IO_L15N_T2_DQS_DOUT_CSO_B_14
T17 IO_L15P_T2_DQS_RDWR_B_14
R17 IO_L16N_T2_D31_14
R16 IO_L16P_T2_CSI_B_14
R20 IO_L17N_T2_D29_14
R19 IO_L17P_T2_D30_14
T20 IO_L18N_T2_D27_14
T19 IO_L18P_T2_D28_14
M_LED_4 AA20 IO_L19N_T3_D25_VREF_14
M_LED_0 Y22 IO_L19P_T3_D26_14
M_LED_1 AA22 IO_L20N_T3_D23_14
M_LED_6 Y20 IO_L20P_T3_D24_14
Y19 IO_L21N_T3_DQS_D22_14
W18 IO_L21P_T3_DQS_14
M_LED_3 AB21 IO_L22N_T3_D20_14
M_LED_2 AA21 IO_L22P_T3_D21_14
M_LED_5 AB20 IO_L23N_T3_D18_14
M_LED_7 AB19 IO_L23P_T3_D19_14
V19 IO_L24N_T3_D16_14
V18 IO_L24P_T3_D17_14
T16 IO_25_14
U3I
XC7S100-2FGGA484I
U15 IO_L1N_T0_13
T14 IO_L1P_T0_13
T13 IO_L2N_T0_13
T12 IO_L2P_T0_13
V15 IO_L3N_T0_DQS_13
V14 IO_L3P_T0_DQS_13
M_HEADER_23 V13 IO_L4N_T0_13
U12 IO_L4P_T0_13
M_HEADER_18 W16 IO_L5N_T0_13
M_HEADER_17 W15 IO_L5P_T0_13
V16 IO_L6N_T0_VREF_13
U16 IO_L6P_T0_13
M_HEADER_9 AA18 IO_L7N_T1_13
M_HEADER_21 Y18 IO_L7P_T1_13
M_HEADER_8 AB18 IO_L8N_T1_13
M_HEADER_7 AA17 IO_L8P_T1_13
M_HEADER_5 AA16 IO_L9N_T1_DQS_13
M_HEADER_3 AA15 IO_L9P_T1_DQS_13
M_HEADER_6 AB17 IO_L10N_T1_13
M_HEADER_4 AB16 IO_L10P_T1_13
M_HEADER_19 Y17 IO_L11N_T1_SRCC_13
M_HEADER_20 W17 IO_L11P_T1_SRCC_13
M_HEADER_2 AB14 IO_L12N_T1_MRCC_13
M_HEADER_1 AA14 IO_L12P_T1_MRCC_13
M_HEADER_13 W12 IO_L13N_T2_MRCC_13
M_HEADER_22 V12 IO_L13P_T2_MRCC_13
M_HEADER_15 Y14 IO_L14N_T2_SRCC_13
M_HEADER_16 W14 IO_L14P_T2_SRCC_13
M_HEADER_0 AB13 IO_L15N_T2_DQS_13
M_TACTSW_3 AA13 IO_L15P_T2_DQS_13
M_HEADER_10 Y11 IO_L16N_T2_13
M_HEADER_11 W11 IO_L16P_T2_13
M_TACTSW_2 AB12 IO_L17N_T2_13
M_TACTSW_0 AB11 IO_L17P_T2_13
M_HEADER_14 Y13 IO_L18N_T2_13
M_HEADER_12 Y12 IO_L18P_T2_13
M_TACTSW_1 AA11 IO_L19N_T3_VREF_13
T15 IO_0_13
U3L
XC7S100-2FGGA484I
F11 IO_L1N_T0_16
G11 IO_L1P_T0_16
F10 IO_L2N_T0_16
G10 IO_L2P_T0_16
D11 IO_L3N_T0_DQS_16
E11 IO_L3P_T0_DQS_16
F14 IO_L4N_T0_16
F13 IO_L4P_T0_16
E12 IO_L5N_T0_16
F12 IO_L5P_T0_16
F15 IO_L6N_T0_VREF_16
G14 IO_L6P_T0_16
C13 IO_L7N_T1_16
D13 IO_L7P_T1_16
A12 IO_L8N_T1_16
A11 IO_L8P_T1_16
A13 IO_L9N_T1_DQS_16
B13 IO_L9P_T1_DQS_16
C11 IO_L10N_T1_16
C10 IO_L10P_T1_16
D15 IO_L11N_T1_SRCC_16
D14 IO_L11P_T1_SRCC_16
C12 IO_L12N_T1_MRCC_16
D12 IO_L12P_T1_MRCC_16
C16 IO_L13N_T2_MRCC_16
C15 IO_L13P_T2_MRCC_16
B16 IO_L14N_T2_SRCC_16
B15 IO_L14P_T2_SRCC_16
C17 IO_L15N_T2_DQS_16
D16 IO_L15P_T2_DQS_16
E16 IO_L16N_T2_16
E15 IO_L16P_T2_16
A14 IO_L17N_T2_16
B14 IO_L17P_T2_16
A17 IO_L18N_T2_16
A16 IO_L18P_T2_16
C18 IO_L19N_T3_VREF_16
D18 IO_L19P_T3_16
B19 IO_L20N_T3_16
C19 IO_L20P_T3_16
A19 IO_L21N_T3_DQS_16
A18 IO_L21P_T3_DQS_16
E18 IO_L22N_T3_16
E17 IO_L22P_T3_16
A20 IO_L23N_T3_16
B20 IO_L23P_T3_16
D19 IO_L24N_T3_16
E19 IO_L24P_T3_16
G13 IO_0_16
G15 IO_25_16
Sheet: /Main FPGA IO/
File: MainFPGAIO.kicad_sch
Title:
Size: A3
Date:
Rev:
KiCad E.D.A. 9.0.1
Id: 17/20



Sheet: /Main FPGA Supervisor/		
File: MainFPGASupervisor.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.1		Id: 18/20



Sheet: /Control FPGA Supervisor/		
File: ControlFPGASv.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.1		Id: 19/20



Sheet: /CurrentSensAmplifer/
File: CurrentSensAmplifer.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.1		Id: 20/20